MONASH University

FACULTY OF INFORMATION TECHNOLOGY

FIT1047 Introduction to computer systems, networks and security – S2 2025

Assignment 3 – Networks

Purpose	In this assignment, students will record data from a real-world wireless network and demonstrate that they can analyse it, identify its properties and potential issues. Also, students will analyse Internet traffic and identify servers, clients and protocols used. The assignment is related to Unit Learning Outcomes 5 and 6.
Your task	**Part 1: Submit your reflections (Week 7, 8 and 9).** **Part 2: Submit a report with your findings regarding the analysis tasks.** **Part 3: In-class in-person test.** (Week 12 Applied [Australia] / Workshop [Malaysia] session).
Value	**30**% of your total marks for the unit. **(10% for Part 2 and 20% for Part 3)** The assignment is marked out of 60 marks.
Word Limit	See individual instructions.
Due Date	**Parts 1 - 2: 11:55 PM Thursday 25 September 2025** ***(NOT Friday 26 September which is a public and Monash holiday)*** **Part 3: Week 12 (Your Officially Allocated Applied Session [Australian cohort] / Allocated Workshop Session [Malaysian cohort])**
Submission	• Via Moodle Assignment Submission. • **DRAFT** upload confirmation email from Turnitin is not a submission. You must click the submit button to accept terms and conditions in Moodle. Note that **DRAFT submissions will not be assessed**. • Once the submission is confirmed, **any requests to revert it back to DRAFT** for resubmission will NOT be accepted. Also, **any incorrect, corrupted, empty or wrong file type** submission will not be assessed. Please check carefully before confirming your submission. • Turnitin and MOSS will be used for similarity checking of all submissions. • This is an individual assignment (group work is not permitted). • In this assessment, you must **not** use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.
Assessment Criteria	Please refer to the rubric at the bottom of Part 2 submission page (https://learning.monash.edu/mod/assign/view.php?id=4434960).

Late Penalties	● 5% deduction per calendar day is automatically applied on Moodle or part thereof for up to one week ● Submissions more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided. ● Late penalty is automatically calculated based on Moodle submission.
Support Resources	See Moodle Assessment page
Feedback	Feedback will be provided on student work via: ● general cohort performance ● specific student feedback ten working days post submission

INSTRUCTIONS

This assignment has THREE parts. Make sure you read the instructions carefully.

For **Part 1**, collect your **reflections** for weeks 7, 8 and 9 from each week's Ed Lesson and create a single PDF document. You can simply copy/paste your reflections, but please add headings for each week. A **template is available** on Moodle. **Submit your PDF through the Moodle Assignment 3 Part 1 activity.**

For **Part 2**, you need to submit the **survey results and report** in a single **PDF** file **through the Moodle Assignment 3 Part 2 activity.**

Part 3 is an **in-class test** during your allocated Applied [Australia] / Workshop [Malaysia] Class in Week 12.

How are marks and grades determined?

Part 2 and Part 3 are worth 20 and 40 marks, respectively. The overall mark is the sum of the two individual marks. The assignment is worth 30% of the unit's marks.
If no meaningful/insufficient/irrelevant reflections are submitted for Part 1, the overall mark will be the maximum of 30 and the sum of Part 2 and Part 3 (i.e., the marks are then capped at 30). For example, if the overall combined mark is 31/60, it will be scaled to 30/60. If the overall combined mark is 28/60 then it will remain as 28/60.

Part 1: Reflection (Not marked, but cap on overall mark applies)

Complete your reflection activities for Week 7 to Week 9 in the corresponding Ed Lesson and copy/paste them into a PDF file. Write at least 100 words for each week (relevant and meaningful to the specific week).

Failure to submit all relevant reflections (missing all submissions or incomplete submissions) will result in your Assignment 3 having a maximum mark of 50%. For example, if the overall combined mark is 31/60, it will be scaled to 30/60. If the overall combined mark is 28/60 then it will remain as 28/60.

You may use this template:
https://docs.google.com/document/d/18UIEJQeyarYW1pl8oDEaf--ubCdJ5LDf-9_jSLbGxrE/edit?usp=sharing to write down your reflections.

Part 2: WLAN Network Design and Security (20 marks)

For this part of the assignment, you will perform a real-world WLAN site survey. Your task is to produce a map of part of a building that gives an overview of the wireless networks that are available, as well as an analysis of the network.

What you will need: a WiFi-enabled laptop (some smartphones also work, see below), and a place to scan. You have to perform a survey of parts of the Monash Clayton / Kuala Lumpur campus.

You have to complete **two tasks** (a survey and a report).

Task 2.1: Survey (6 marks)

For Australian campus cohort:
Create a map and survey of (a part of) the following building on our Clayton Campus:

- Students with student number ending with “1” or “6”: Woodside Building
- Students with student number ending with “2” or “7”: Hargrave Andrew Library
- Students with student number ending with “3” or “8”: Sir Louis Matheson Library
- Students with student number ending with “4” or “9”: Learning and Teaching Building
- Students with student number ending with “5” or “0”: Menzies Building

(You can find the location of the building here - Choosing Clayton campus)

For Malaysian campus cohort:
Create a map and survey of (a part of) any building of your own choice on our Malaysian Campus.

(You can find the location of the building here - Choosing Kuala Lumpur campus)

1. Draw a floor plan with details

A simple floor plan will be sufficient. It does not have to be perfectly to scale. See Appendix A for an example. The map should be labelled with all relevant information (e.g., dimension, door, wall and material such as wood or concrete or glass, if used for the discussion). Your survey should cover an area of at least 60 square metres (e.g., 6x10 metres, or 4x15, or two storeys of 6x5 each). Be sure to take the analysis in Task 2.2 into account, by designing your survey to include walls, doors etc. it will be easier to write something interesting in Task 2.2.

For drawing the site maps, any drawing tool should work, for example LucidChart, or even presentation tools such as PowerPoint, Keynote or Google Slides. Modification of screen capture map (e.g., from Monash Digital Map) is acceptable provided that the appropriate references are provided or marks will be deducted. Use the APA 7th referencing style. Scans of hand-drawn maps are ***not*** acceptable.

An example map is given in the Appendix A. You may use any drawing tool to create a map (excluding heatmap generated by survey tools) or reuse existing floor plans with reference.

2. Conduct the survey

Your survey must include at least three WiFi access points. If you want, you can create an additional AP with your phone (using "Personal hotspot" or "Tethering" features).

For the survey, use a WLAN sniffing tool (see below) in at least eight different locations on your map. For each location, record the technical characteristics of all visible APs. Depending on the scanning tool you use, you record features such as the network name, MAC address, signal strength, 802.11 version(s) supported, band (2.4/5/6 GHz), channel(s) and security used.

Take screenshots of survey data at each survey location, and include the screenshots of raw data in the Appendix of your report.

Tools: You can use NetSpot (http://www.netspotapp.com) for macOS and Windows, LinSSID or wavemon for Linux. Acrylic WiFi (https://www.acrylicwifi.com/en/) is another possible choice for Windows. If you have an Android smartphone, apps like Wifi Analyzer (https://play.google.com/store/apps/details?id=abdelrahman.wifianalyzerpro) can also be used. On iOS, WiFi scanning apps do not provide enough detail, so iPhones won't be suitable for this task.

3. Add the data into the map

Add the gathered data from the survey into the map of the covered area. On the map, indicate the location of the access points and the locations where you took measurements.

For the access points, use the actual location if you know it, or an approximation based on the observed signal strength (e.g. if you don't know exactly where it is).

For each measurement point, you either add the characteristics directly into the map using any annotation feature/tool, or create a separate table with the details. You can submit several maps if you choose to enter data directly into the maps, or a single map if you use additional tables. Create the map yourself, do not use the heat-map mapping features available in some commercial (i.e., paid) WiFi sniffing tools.

Task 2.2: Report (14 marks)

Write a report (word limit: no more than 1000 words) on your observations analysing the data collected in the previous step (Task 2.1). Your analysis must investigate the following aspects:

1. **Channel allocation and overlap: (2 marks)**
    - Are the access points using overlapping or adjacent channels?
    - Is there evidence of co-channel interference?
    - Suggest improvements to channel planning.
2. **Signal attenuation and obstacles: (2 marks)**
    - How do physical barriers such as walls, glass, or furniture affect the signal?
    - Compare at least two different material types in terms of attenuation.
3. **Coverage and dead zones: (2 marks)**
    - Identify areas with weak or no connectivity.
    - Recommend changes in AP placement, orientation, or transmit power to improve coverage.
4. **Roaming and handoff potential: (2 marks)**
    - Is there sufficient overlap between APs to allow seamless roaming?
    - How might roaming performance impact user experience (e.g., VoIP calls, video streaming)?
5. **Network load and bandwidth usage: (2 marks)**
    - Do multiple APs appear to be heavily utilised?
    - Suggest strategies to balance load or optimise performance.
6. **Other analysis of your choice (1 or more): (2 marks)**
    - For example:
        - Estimate AP locations from observed signal strength.
        - Test body attenuation.
        - Consider interference from non-WiFi devices (e.g., microwaves, Bluetooth).
    - Explain both your methodology and reasoning.

Report structure: (2 marks)

Task 2.2 requires you to write a structured report, not just a series of answers to the above questions. Ensure that your discussion flows logically, with each section connected and presented clearly (e.g., using sub-header to indicate the beginning of each section), rather than presenting unrelated short answers, or writing a single paragraph for the whole task and putting everything into it. Figures, tables, and references to your survey data may be included to support your analysis.

Part 3: Quiz - Internet Traffic Analysis and Basic Cryptography (40 marks)

This part of the assignment must be done <u>in-person</u> during your Week 12 allocated Applied Session (Australian Cohort) / Workshop Session (Malaysian Cohort). You will be given <u>one hour</u> to complete this part. You cannot start this part elsewhere or in another time slot. Bring your student ID and own device.

There are two sections for Part 3:

Section 1: The first section of the quiz requires you to download a PCAP file, open it in Wireshark and answer a few questions about the captured frames. The PCAP files are individualised, so make sure that you download the correct file while you are logged into Moodle.

You can access your individual PCAP file through the <u>Assignment 3 Part 3 In-class test</u> link on Moodle at your Week 12 Applied (Australia) / Workshop (Malaysia) session. All of your answers have to be submitted via Moodle.

Here are a few tips on how to approach these tasks.

Node Selection:
Please make sure you select the correct node (within your given scenario) for traffic analysis.

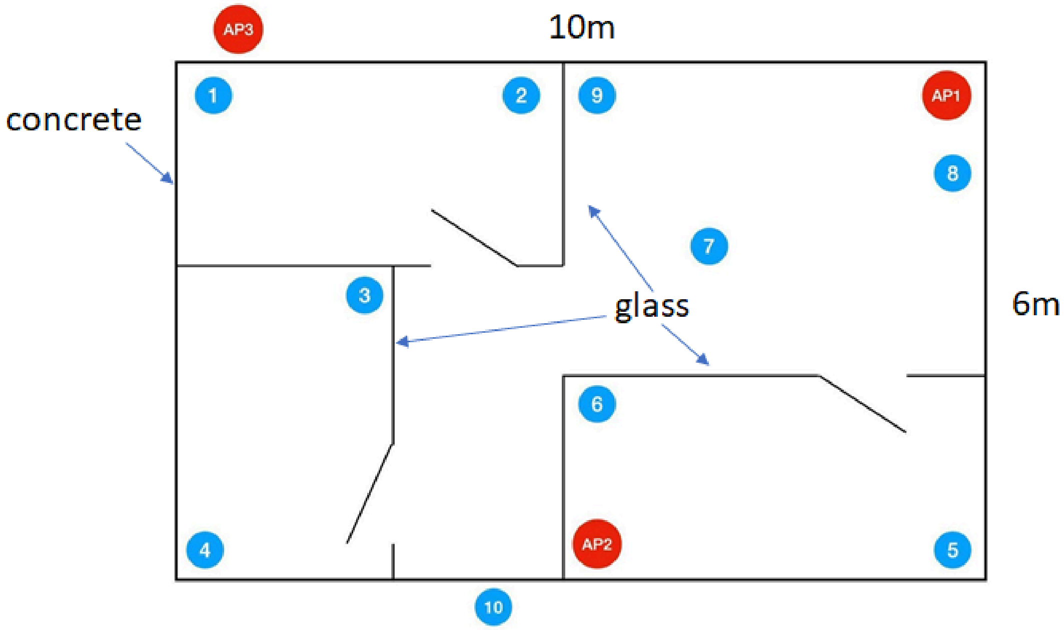
MAC addresses:
These are the addresses of individual devices at the Data Link Layer. Each frame contains a sender and receiver MAC address. For each frame, think about which device would be the sender and which the receiver.

IP addresses:
These are the Network Layer addresses. Remember that we use the DNS protocol to map a human-readable address (such as www.monash.edu) to an IP address (such as 202.9.95.188). So in order to find out the IP address for some of the devices, you may have to try to find DNS requests and responses in the PCAP file.

TCP connections:
Remember that each TCP connection starts with a three-way handshake. This was covered in the lectures, so you may have to go back to the videos if you're not sure what those frames look like.

Section 2: The second section is about some basic questions of cryptography. More details will be given to you in Week 10 and Week 11.

Appendix A: Sample map only showing APs and survey locations



Building ABC

Remarks: Note that even if you cannot enter a classroom (e.g., G02 and G03 below), you can still take the reading inside (e.g., 1, 2) and outside the building (e.g., 3, 4, 5) to conduct the survey.